Question Paper

Exam Date & Time: 28-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/DEC 2024

MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-III [MAT 3135]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Missing data may be suitably assumed

1) A) (i) Find all solutions, if any, of the linear congruence $20x \equiv 4\ (mod\ 30)$. (3)

(ii) What is the value of the sum

$$\phi(1) + \phi(3) + \phi(3^2) + \cdots + \phi(3^{n-1}) + \phi(3^n)?$$

Justify your answer with relevant theorems.

B) Find the least positive integer $x > 1$ that satisfies the following congruences simultaneously: (3)

$$x \equiv 1(mod\ 3), x \equiv 1\ (mod\ 5), x \equiv 1\ (mod\ 7).$$

C) (i) Note that $\{1, 2,4,5,7,...,19,20\}$ forms a reduced residue system modulo 21. Give any integer value of k greater than 1 for which the set $\{1, 2^k, 4^k,..., 20^k\}$ is again a reduced residue class modulo 21 and justify your choice of k. (4)

(ii) Are the two binary quadratic forms $x^2 + xy + 2y^2$ and $x^2 - xy + 2y^2$ equivalent? Justify.

2) A) Does there exist a solution for the quadratic congruence $x^2 \equiv -2\ (mod\ 61)$? Justify your answer. If so, how many? (3)

B) Find all the primes p such that $\left(\frac{5}{p}\right) = -1.$ Justify your answer. (3)

C) (i) What is the highest power of 3 dividing 533!? (4)

(ii) Find the value of $2^{45}\ (mod\ 91)$ using repeated squaring.

3) Alice wants to secretly send a one-letter alphabet using public key cryptographic techniques to Bob. Using ASCII encoding, she converts the alphabet into a natural number a which is co-prime to 85 (3)

A) and encrypts it using the public key $(k, m) = (5,85)$. Let b denote the number obtained after her encryption.

(i) What is the key that Bob should use to decrypt it? In other words, find a positive integer h such that $b^h \equiv a \pmod{m}$.

(ii) If the ASCII symbol for b is $ (Dollar sign), what is the original message in plain text sent by Alice?

Note: ASCII tables are provided in the reference for converting symbols/letters to numbers and vice versa.

B) If an odd prime p satisfying $p \equiv 3 \pmod 4$ is represented by the integral binary quadratic form $x^2 + 5y^2$, then what are the possible values of $p \pmod 5$? Justify your answers. (3)

C) In a graph G, if $diameter(G) \geq 3$, then show that $diameter(\overline{G}) \leq 3$. (4)

Hence, show that every non trivial self-complementary graph has diameter 2 or 3.

4) If G is a graph with n vertices and minimum degree $\delta(G) \geq \frac{n-1}{2}$, then prove that G is (3)

A) connected.
Give an example for a graph on 6 vertices with minimum degree 3.

B) Prove that a tree T on n vertices has n-1 edges. (3)
Draw a tree with 8 vertices having exactly 6 pendant vertices.

C) Obtain the betweenness centrality of all the vertices in (4)

(i) a cycle graph on 8 vertices

(ii) a complete bipartite graph, $K_{1, n-1}$

(iii) a path graph on 5 vertices

5) Let G be a labelled graph with A(G) or A as its adjacency matrix. Prove that the $(i,j)^{th}$ entry of A^k is the number of walks of length k from the vertices v_i to v_k. (3)

A) Wrire the adjacency matrix A of complete graph K_5 and find the $(1,5)^{th}$ entry of A^3 .

B) Obtain the determinant of the adjacency matrix of (3)

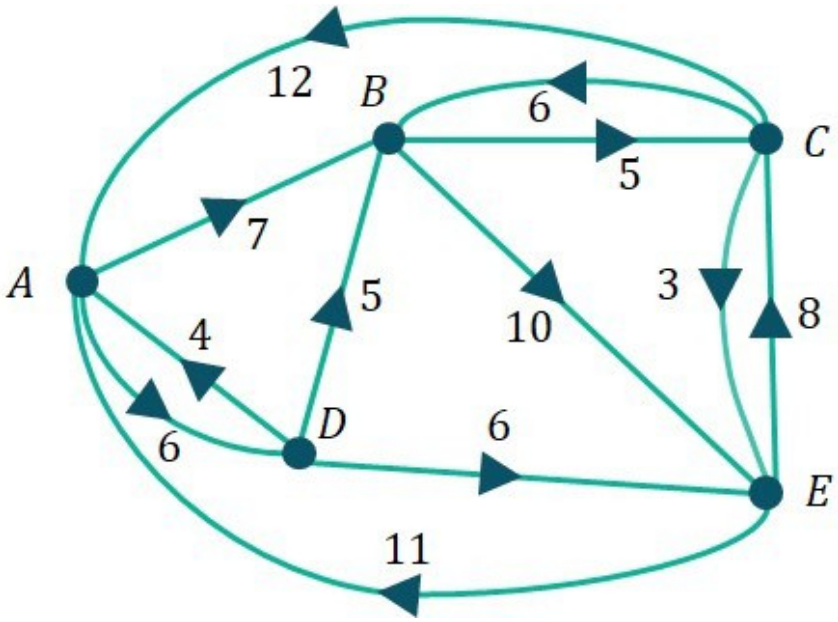
(i) a cycle graph on n vertices,

(ii) a path graph on n vertices,

(iii) a complete graph on 4 vertices,

by obtaining all the elementary spanning subgraphs and using the formula $\det(A(G)) = \sum(-1)^{n-c_1(H)-c(H)}\, 2^{c(H)}$, where summation runs over all elementary spanning subgraphs H of G and $c_1(H)$ and $c(H)$ are the number of components which are $K_2's$ and cycles respectively.

C) Using Dijkstra's algorithm, obtain the shortest path from the vertex B to every other vertex for the graph as shown below (4)



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